

Single Crystal Growth and Characterization of NdFeO₃: A Novel Candidate for Topological Magnons



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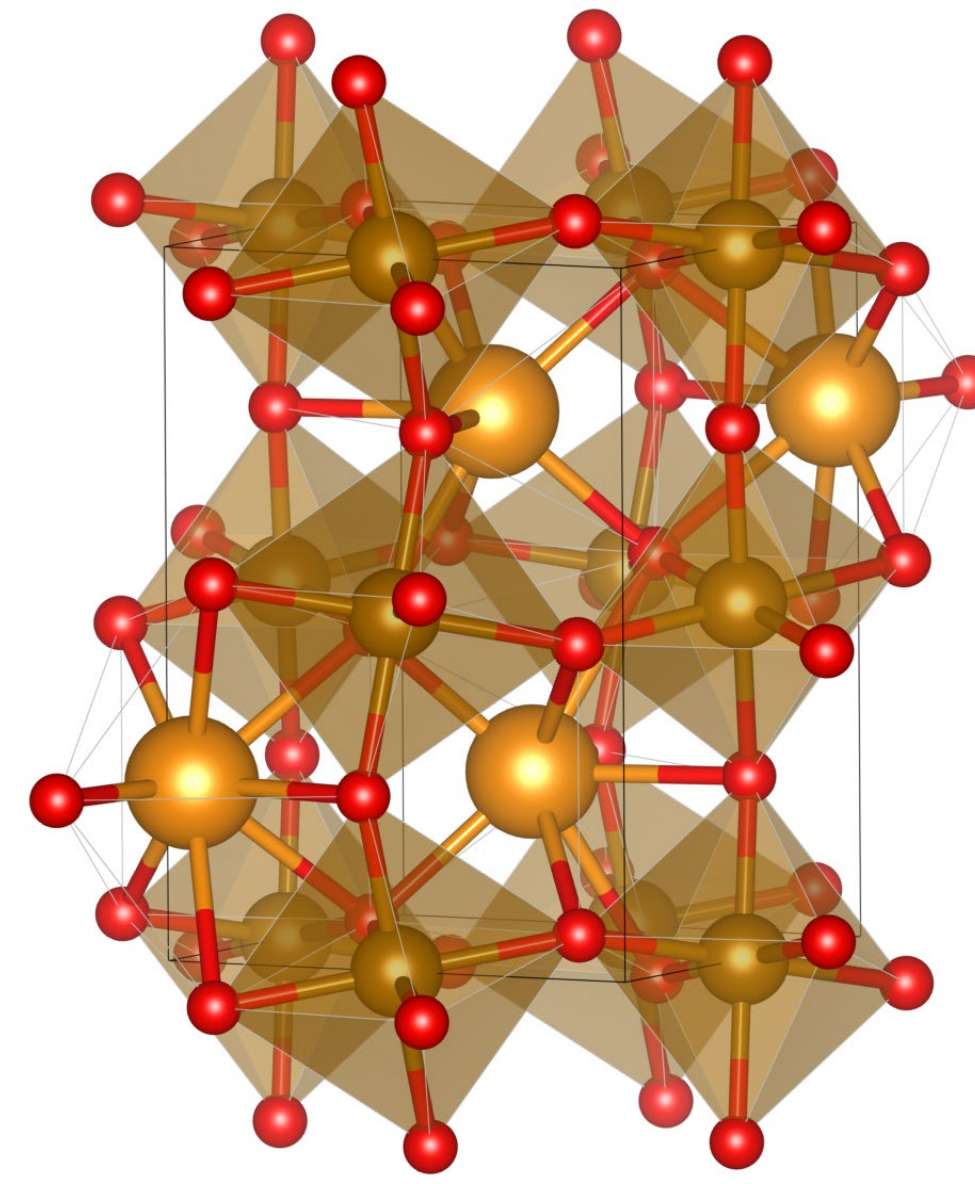
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Introduction: Topological Magnons & NdFeO₃

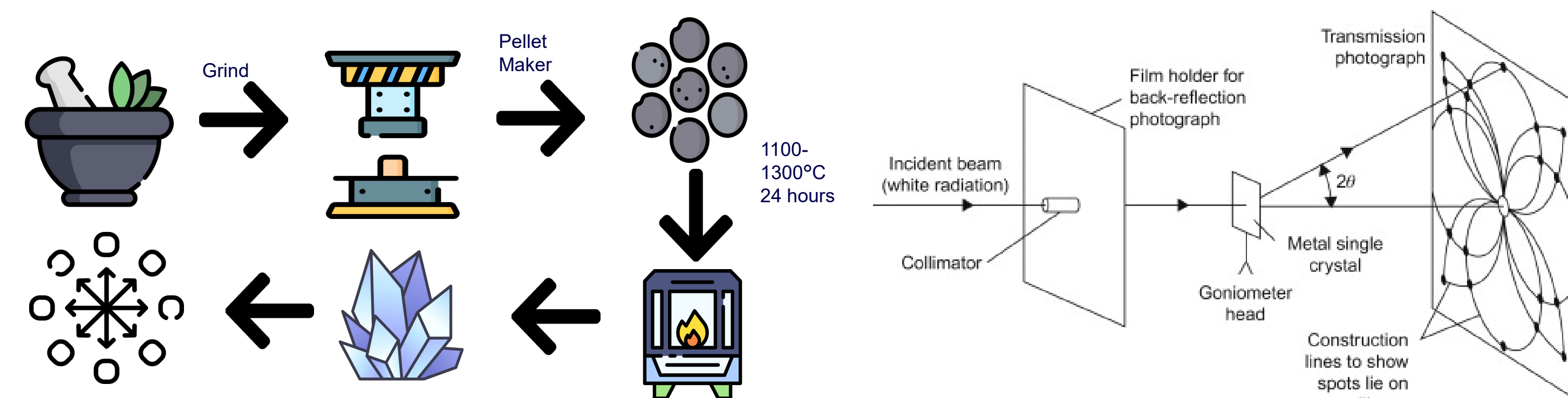
- Topological Magnons are collective spin excitations in magnetic materials with non-trivial band topology.
- Such magnonic edge states enable low-dissipation spin transport for energy-efficient computing and quantum devices, especially in spintronics.
- NdFeO₃ crystallizes in an orthorhombic *Pnma* perovskite structure with a Néel temperature of 760 K.
- It features Nd³⁺ at cube-corner sites, Fe³⁺ in octahedral centers, and O²⁻ in apical/equatorial positions of tilted FeO₆ octahedra with *a*=3.51, *b*=6.25, *c*=13.06 angstroms.
- We grow a single crystal of the topological magnon candidate NdFeO₃ and report some properties.



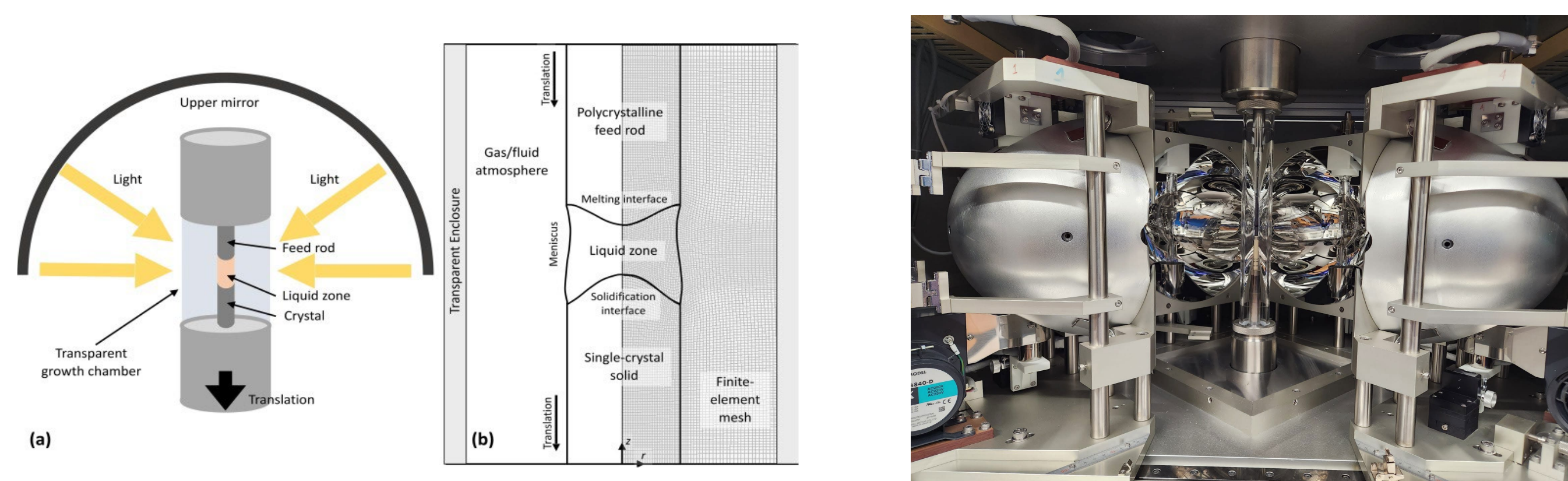
NdFeO₃ Crystal Structure

Methods: Crystal Growth and Characterization

Solid State Synthesis, Diffraction, and Laue Scattering



Optical Floating Zone Furnace



Quantum Design PPMS



Results, Analysis, and Discussions

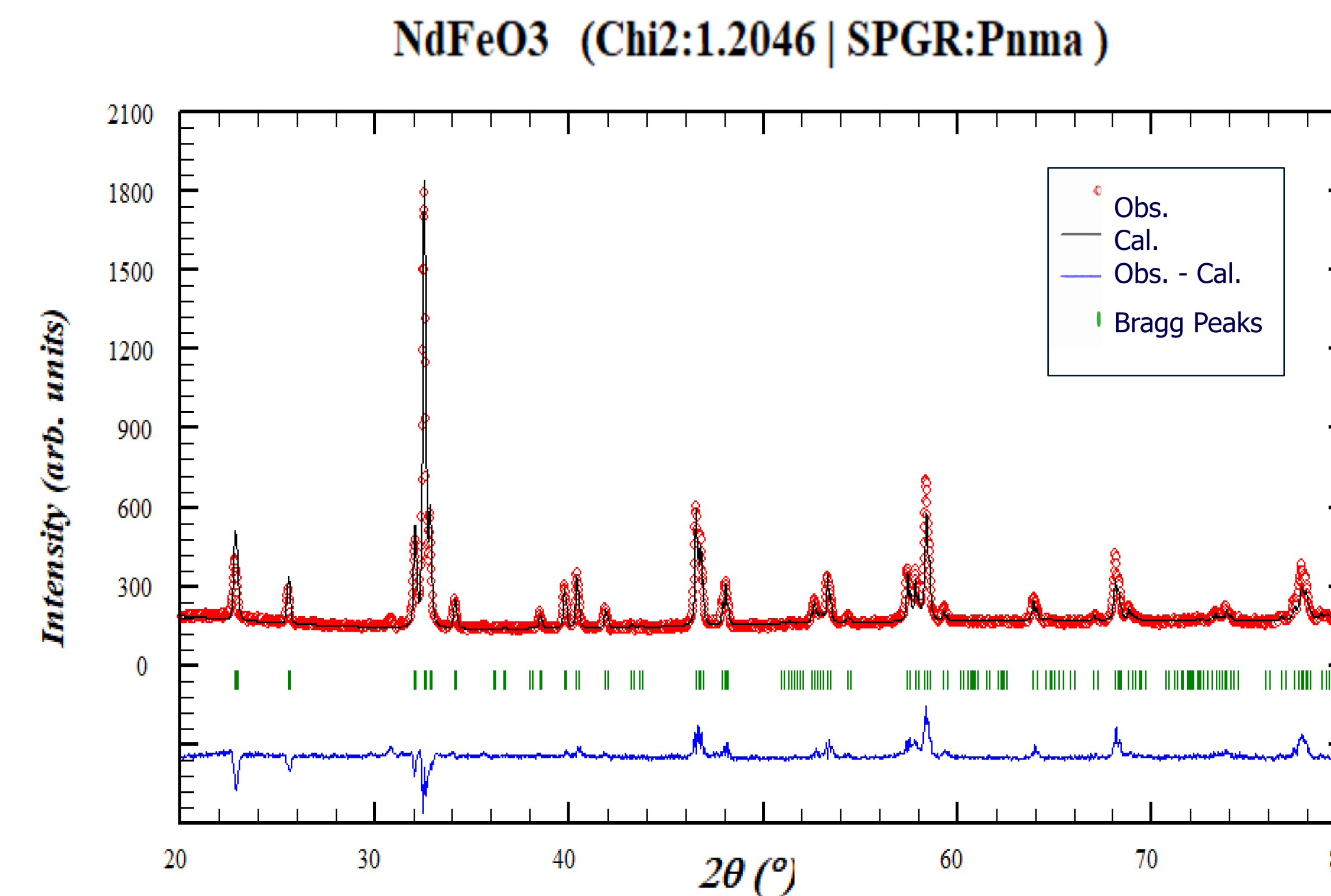


Figure: Rietveld Refinement Pattern of the Synthesized NdFeO₃.

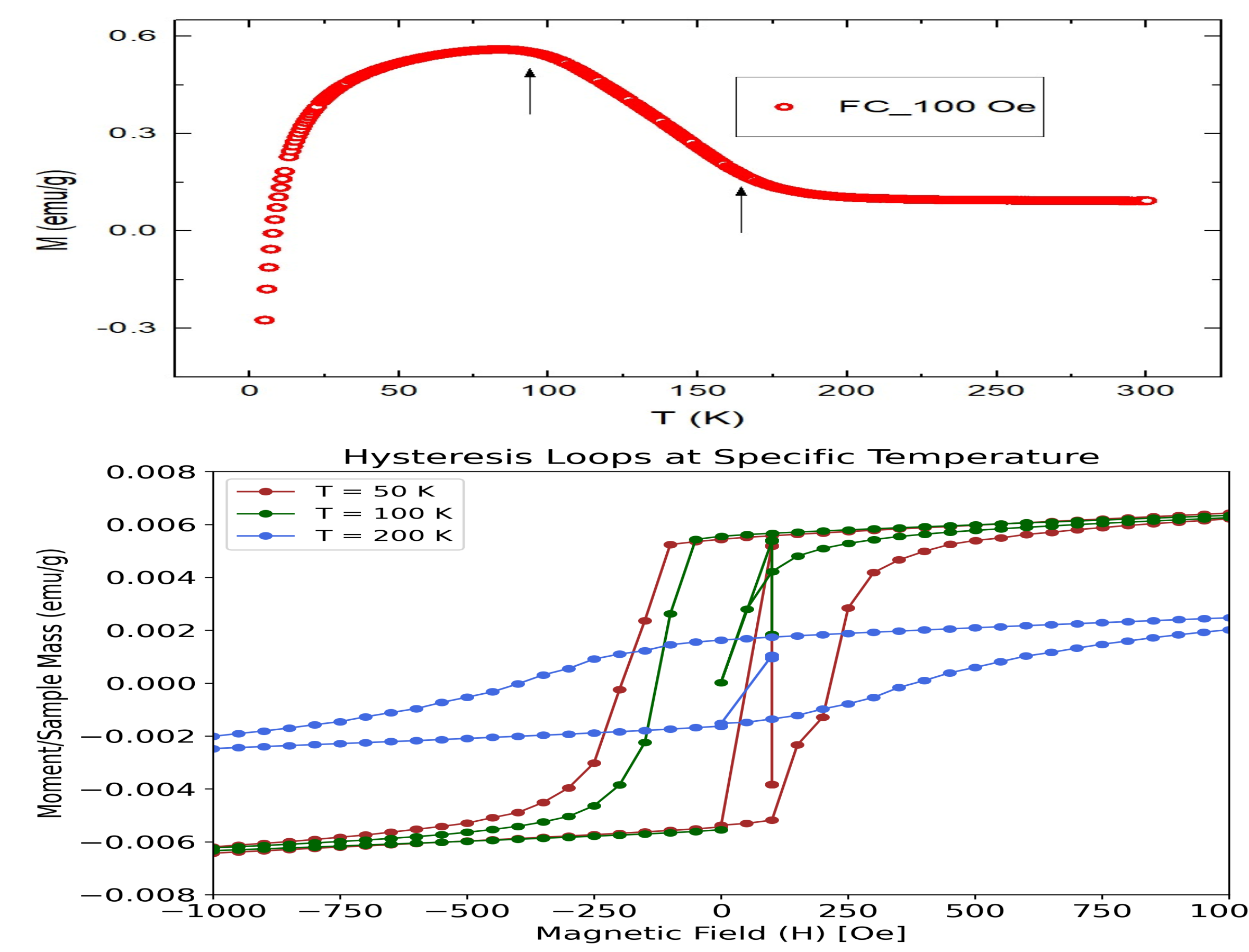


Figure: Temperature Dependence of Magnetization and Hysteresis Loops in NdFeO₃.

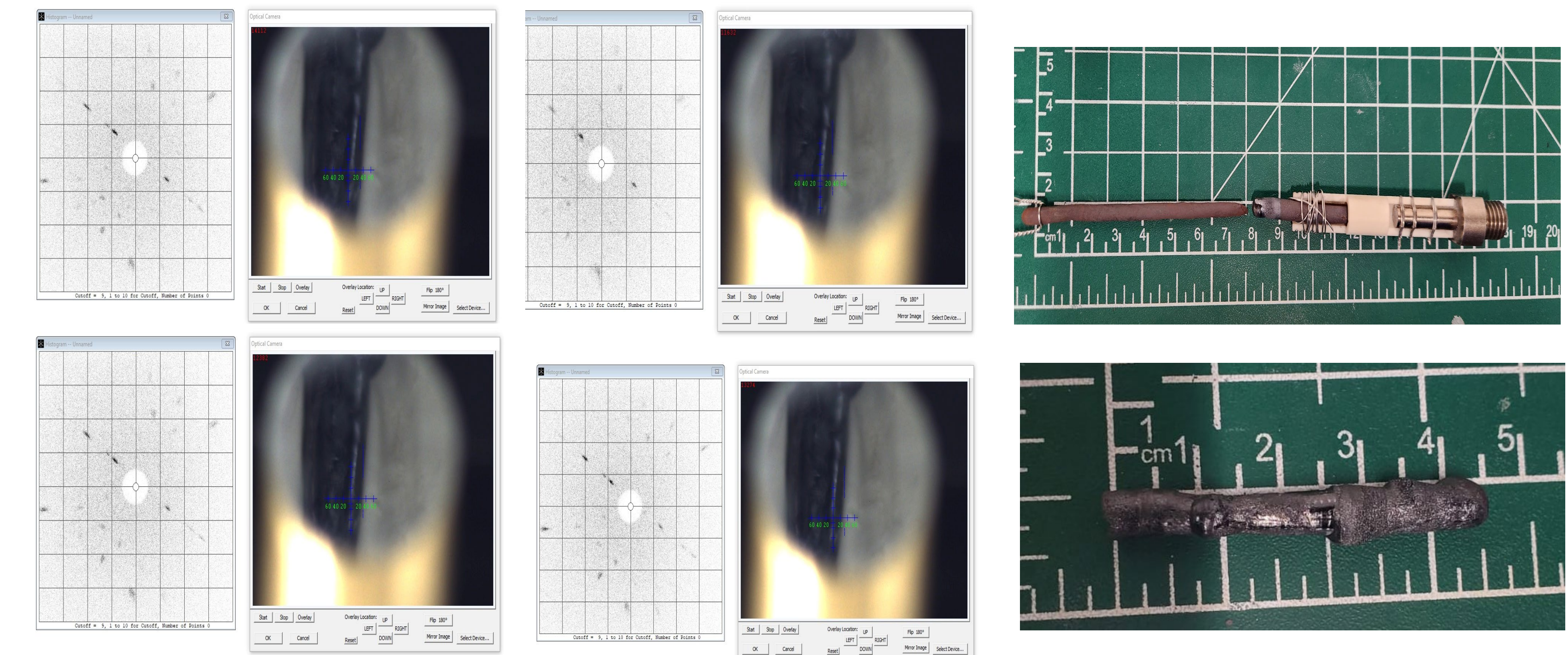


Figure: Captured Images Showing Laue Patterns (left) and Rods Before and After Floating Zone Growth (right).

Results and Discussions

- A pure sample of NdFeO₃ was synthesized and a single crystal was grown.
- Through XRD and Laue Scattering, the crystal's purity and formation was checked.
- The non-oriented, field cooled (FC) magnetization (*M*) vs temperature (*T*) measurement shows a clear spin reorientation transition through a wide range of temperature range from 100 to 170K.
- All hysteresis loop show typical antiferromagnetic feature after saturation, which is consistent with the weak ferromagnetism nature of the rare-earth orthoferrites below its Néel temperature.
- Karaki et. al. (2023) theoretically predicted that NdFeO₃ is a topological magnon through a search algorithm. Neutron scattering is further required to probe it.

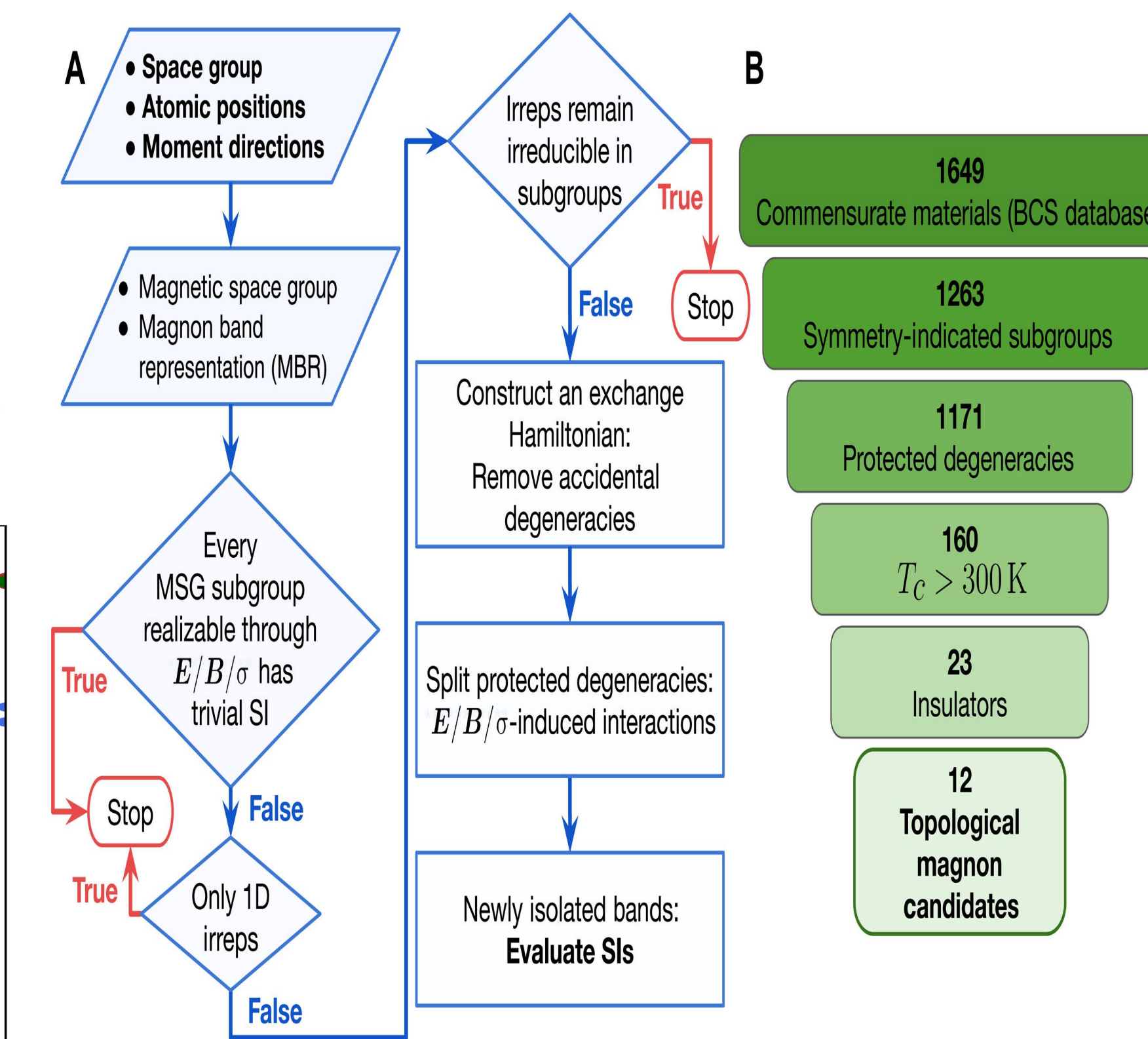


Figure: Flowchart of Topological Magnon Search Algorithm (Adapted from Karaki et. al. 2023).

Conclusions

- A single crystal of NdFeO₃ was grown using optical floating zone technique and its purity was verified using X-ray Diffraction and Laue Scattering.
- M* vs *T* and *M* vs *H* curves were plotted through in-house measurements using SQUID.
- Spin reorientation and antiferromagnetic behaviour were observed through a wide temperature range.
- Neutron scattering is further required to probe the predicted topological magnon behavior.
- Future work will involve re-growth of longer NdFeO₃ crystal and oriented magnetometric measurements with SQUID and PPMS, and finally, with neutron scattering experiment.

References

- Karaki, M. J., Xu, Y., Williams, A. J., Nawwar, M., Doan-Nguyen, V., Goldberger, J. E., & Lü, Y. (2023). An efficient material search for room-temperature topological magnons. *Science Advances*, 9(7). <https://doi.org/10.1126/sciadv.ade7731>
- Shalini, T., Vijayakumar, P., & Kumar, J. (2020). Studies on structural and magnetic properties of NdFeO₃ single crystals grown by optical floating zone technique. *Bulletin of Materials Science*, 43(1). <https://doi.org/10.1007/s12034-020-02259-4>
- Yuan, S., Wang, Y., Shao, M., Chang, F., Kang, B., Isikawa, Y., & Cao, S. (2011). Magnetic properties of NdFeO₃ single crystal in the spin reorientation region. *Journal of Applied Physics*, 109(7). <https://doi.org/10.1063/1.3562259>